Benedikt Seiter

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Education

ETH Zürich Zurich, Switzerland
M.Sc. in Applied Mathematics, focus area: Machine Learning Sep 2026–Spring 2028 (expected)
Current courses: Probabilistic Artificial Intelligence; Natural Language Processing; Advanced Machine Learning.

B.Sc. in Mathematics Sep 2023–Aug 2026
GPA: 5.6/6.0
Thesis: *Anchor PCA* (6.0/6.0), advised by Prof. Jonas Peters.
Key courses: Mathematical Statistics; High-Dimensional Statistics; Computational Statistics; Machine Learning.

Research interests

Domain generalization; foundation models; causality; mechanistic interpretability; LLM evaluation; post-training.

Publications

B. Seiter, A. Fries, J. von Kügelgen, and J. Peters. *Anchor PCA*. NeurIPS 2026 (accepted). arXiv:2606.06233.

Research & projects

Anchor PCA Apr 2026–present
ETH Zürich, Seminar for Statistics PDF GitHub

- Developed Anchor PCA for multi-domain dimensionality reduction, balancing explained variance with stability under domain shift; implemented estimators in Python with NumPy and SciPy.
- Benchmarked against pooled and worst-case PCA in an end-to-end pipeline; improved out-of-domain explained variance on financial and gas sensor data.
- Established recovery of a maximal invariant subspace and proved minimax robustness guarantees.

Transformer Interpretability and Experimentation Ongoing
Independent study through ARENA; implementing activation interventions, linear probing, and sparse autoencoders in PyTorch; running controlled experiments on open-weight LMs using remote GPUs.

Benchmarking Invariant Risk Minimization for Financial Time Series Sep 2025–Jan 2026
ETH Zürich, Supervised Research Project
Reproduced and extended InvariantStock, an IRM-based VAE for regime-robust equity prediction, in PyTorch; rebuilt its pipeline and benchmarked VIX-based environments and loss variants.

Seminar on Representation Learning Sep 2025–Jan 2026
ETH Zürich
Analyzed and presented papers on deep learning, autoencoders, ICA, contrastive and self-supervised learning, causality, and single-cell biology applications.

Teaching

Teaching Assistant Sep 2025–present
ETH Zürich
Lead weekly problem-solving sessions in Theoretical Computer Science and Probability & Statistics.

Awards

First Prize, Bundeswettbewerb Mathematik 2023
Top 2.5% in Germany's national mathematics competition; invited to the German IMO team selection examinations.

DPG-Abitur- und Buchpreis (Physics) 2023
Dr. Hans Riegel-Fachpreis (Mathematics) 2022

Skills

Programming: Python, C++, R, MATLAB **Libraries:** PyTorch, NumPy, SciPy, scikit-learn, pandas
Tools: Git, LaTeX **Languages:** German (native), English (C2, IELTS 8.5)